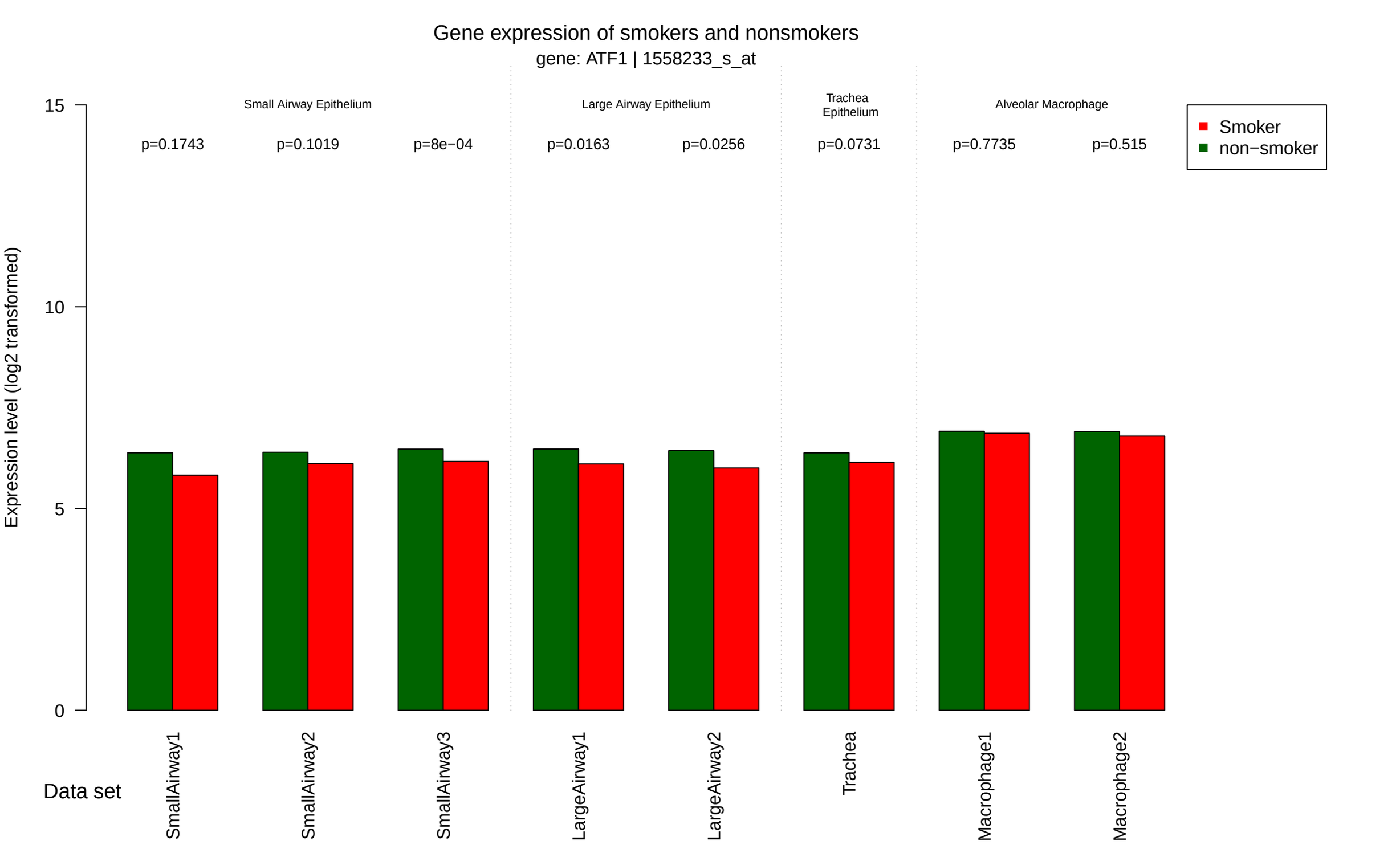
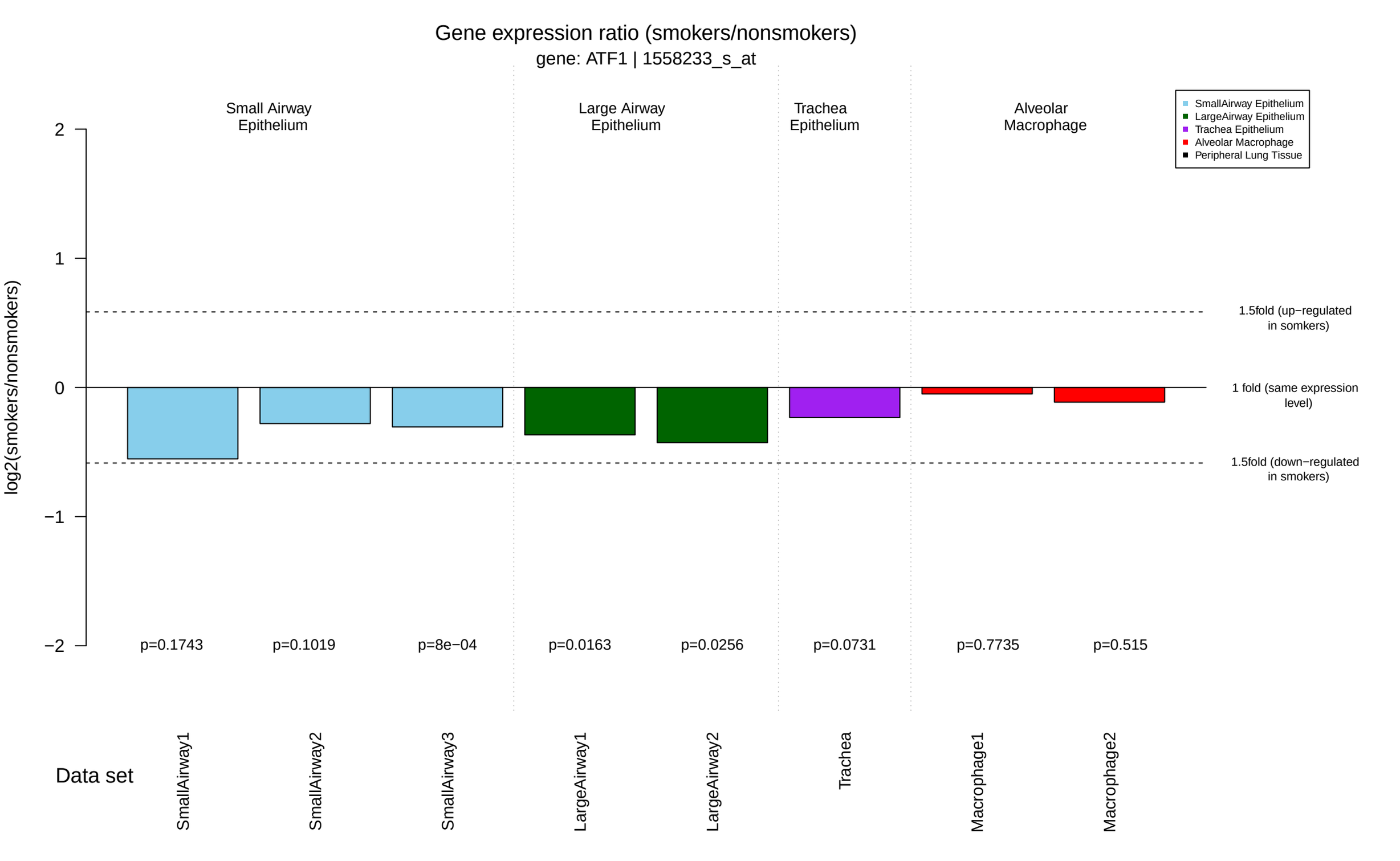
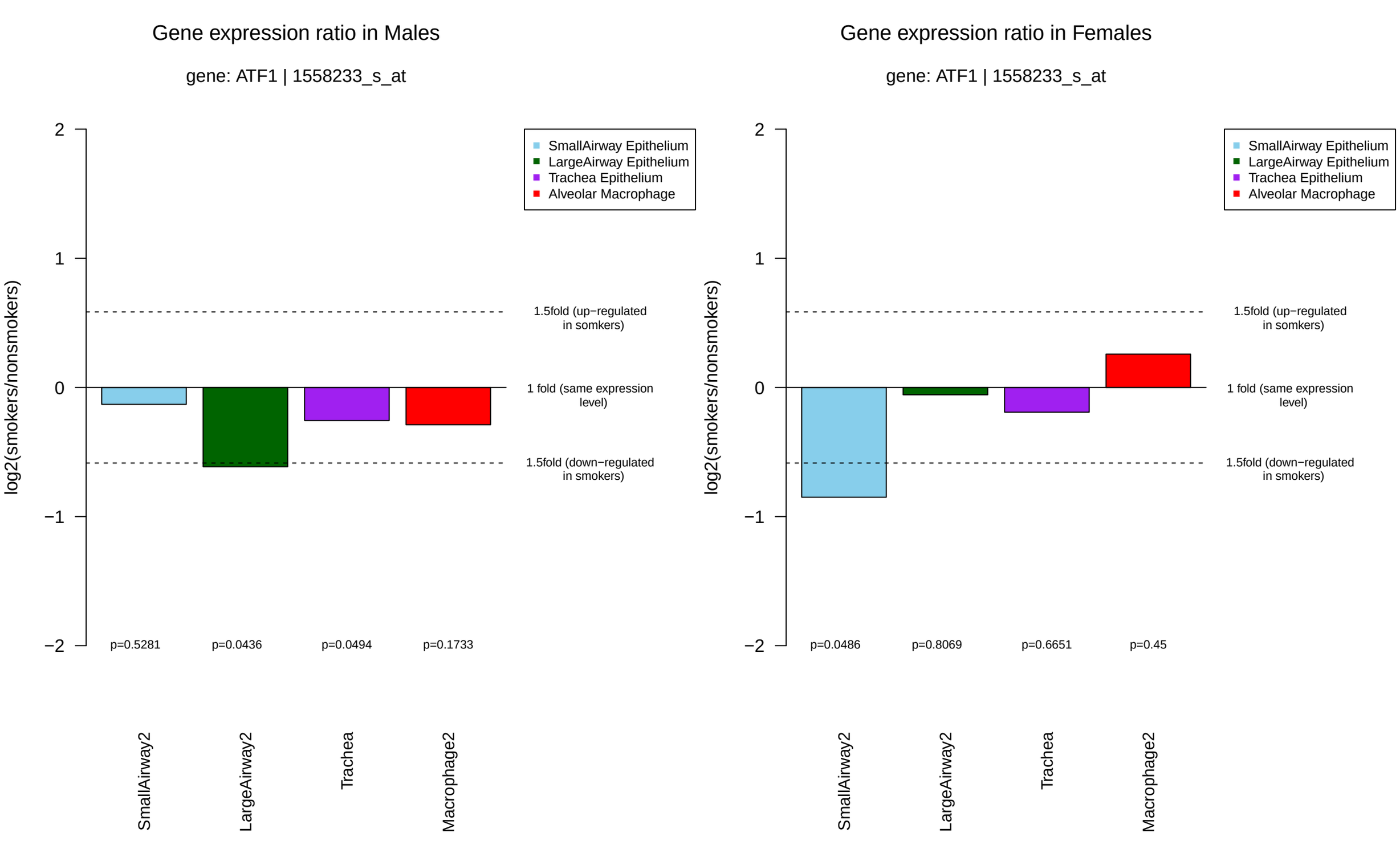
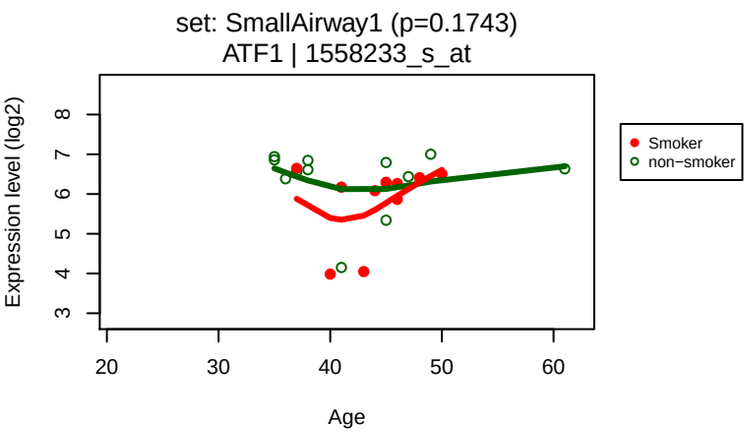
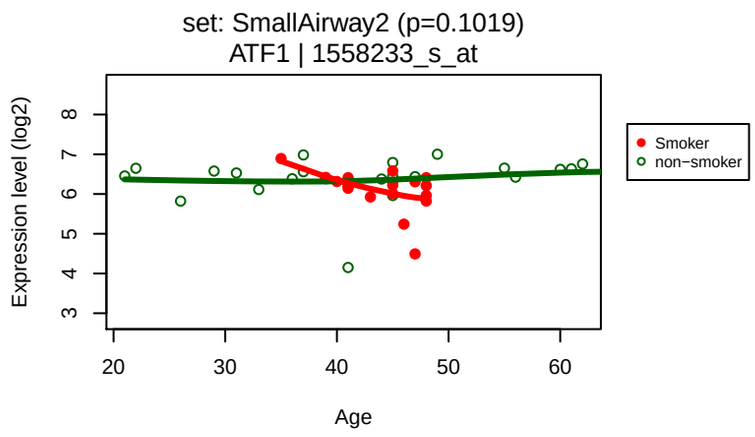
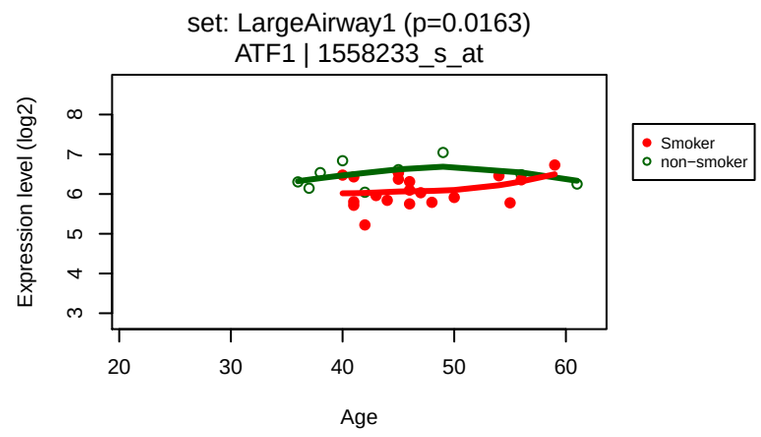

Gene expression of smokers and nonsmokers
gene: ATF1 | 1558233_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1743
p=0.1019
p=8e−04
p=0.0163
p=0.0256
p=0.0731
p=0.7735
p=0.515
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

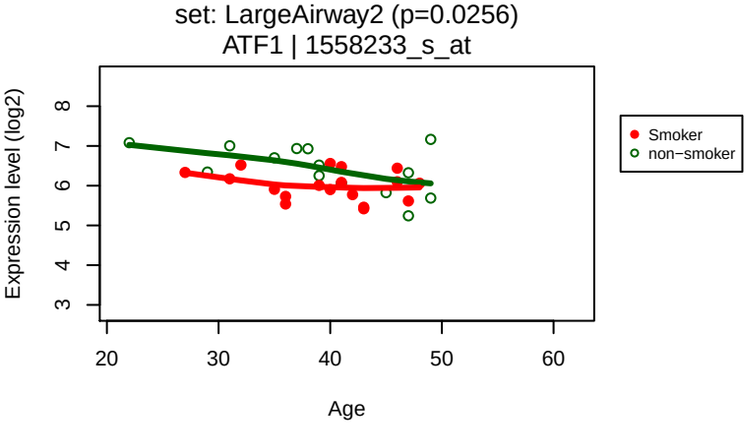

Gene expression ratio (smokers/nonsmokers)
gene: ATF1 | 1558233_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1743
p=0.1019
p=8e−04
p=0.0163
p=0.0256
p=0.0731
p=0.7735
p=0.515
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

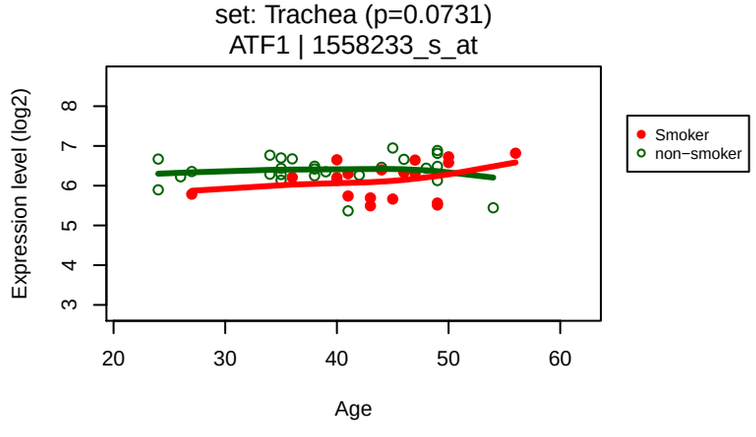

Gene expression ratio in Males
gene: ATF1 | 1558233_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5281
p=0.0436
p=0.0494
p=0.1733
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF1 | 1558233_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0486
p=0.8069
p=0.6651
p=0.45
SmallAirway2
LargeAirway2
Trachea
Macrophage2

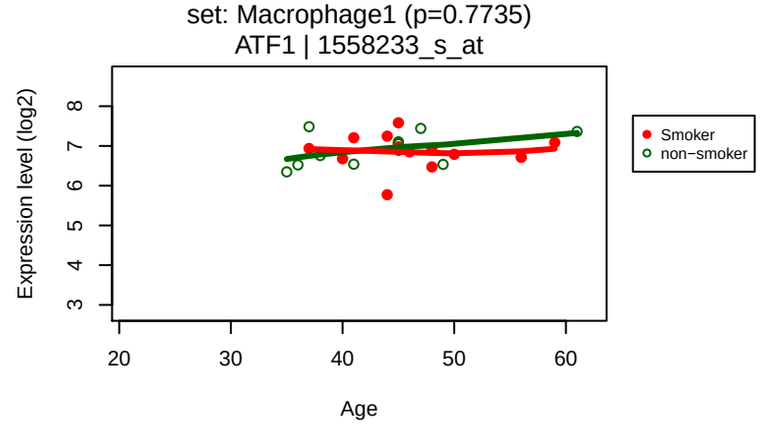

set: SmallAirway1 (p=0.1743)
ATF1 | 1558233_s_at
Expression level (log2)
Age
Smoker
non−smoker

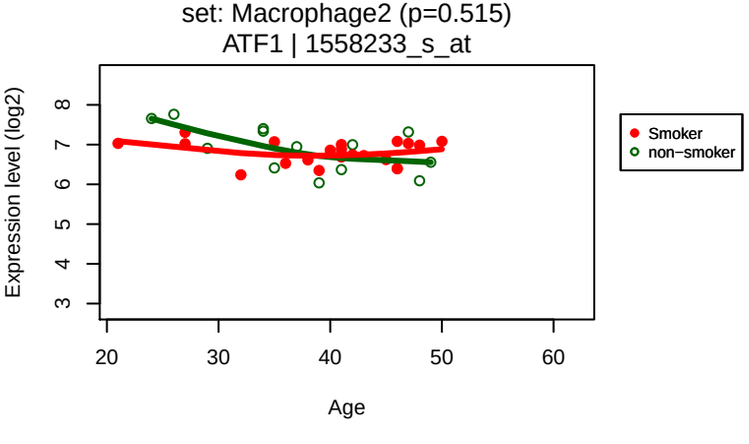

set: SmallAirway2 (p=0.1019)
ATF1 | 1558233_s_at
Expression level (log2)
Age
Smoker
non−smoker

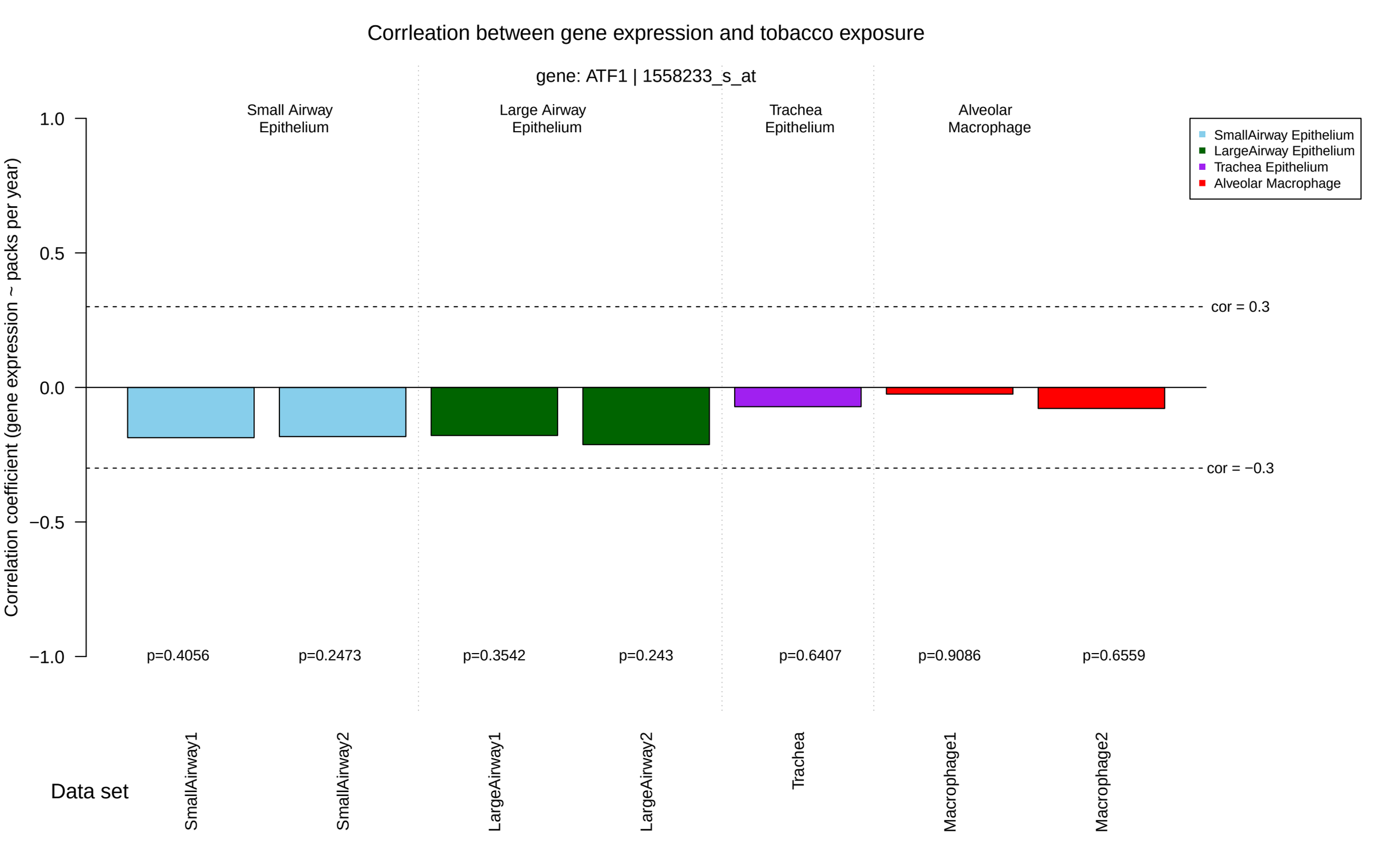

set: LargeAirway1 (p=0.0163)
ATF1 | 1558233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0256)
ATF1 | 1558233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0731)
ATF1 | 1558233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7735)
ATF1 | 1558233_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.515)
ATF1 | 1558233_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF1 | 1558233_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4056
p=0.2473
p=0.3542
p=0.243
p=0.6407
p=0.9086
p=0.6559
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2